

SIH26009 — Complete Teardown

Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls

Organization	Ministry of Steel (MOIL Ltd.)
Theme	Space Technology
Category	Software
Deadline	20 September 2026
Score	21/25 (Tier 1-2 boundary)

1. What MOIL Actually Wants

The PS description is only ~150 words — unusually short. Here's what each line really means:

"Reserve estimation and production planning are mainly based on manual surveys, drilling results, and production records"

Translation: Their geologists walk the field, drill holes, look at samples manually. Slow, expensive, and they often get it wrong.

"mismatch between expected and actual ore production"

Translation: They plan to extract X tonnes from a zone, but actually get Y. This costs them crores in missed targets and idle equipment.

They want AI/ML that uses:

- Geological data — drill logs, rock composition
- Historical production — past output from each mine zone
- Equipment performance — downtime, maintenance cycles
- Satellite inputs — rainfall, soil moisture, vegetation index, land temperature

Three deliverables:

- Map reserves more accurately (where is the manganese?)
- Predict production shortfalls (will we miss target this month?)
- Suggest corrective actions (reschedule, redeploy equipment)

2. The REAL Problem (Not What You Think)

Most teams will read this and think: *"I need to find manganese from space."*

Wrong. Read it again carefully.

The satellite inputs they listed are: rainfall, soil moisture, vegetation index, land temperature. These are **NOT** for mineral detection — these are **environmental factors that affect mining operations**.

MOIL already knows where their manganese is (they've been mining 60+ years). What they DON'T know is:

- Will heavy monsoon rain shut down operations next week?
- Is equipment likely to fail based on usage + weather conditions?
- Should they shift production from Mine A to Mine B this quarter?

The real PS is 70% production prediction + 30% reserve mapping.

This changes everything about feasibility.

3. Feasibility Breakdown

Component	Level	Why
Satellite weather / environmental data	EASY	Sentinel-2, Landsat free data. NDVI, soil moisture, rainfall — all standard products. Google Earth Engine makes this trivial.
Production shortfall prediction	MEDIUM	Time-series forecasting on production + weather + equipment data. Standard ML (XGBoost, LSTM). Simulate historical data if MOIL doesn't share.
Equipment downtime prediction	MEDIUM	Predictive maintenance is a well-solved ML problem. Sensor data to failure prediction. Plenty of reference implementations.
Reserve mapping from satellite	HARD	Published research exists on Dongri Buzurg mine (MOIL's own!) using Landsat 9 multispectral. Not impossible but needs domain prep.
Corrective action suggestions	EASY	Rule-based system on top of predictions. "If rain > threshold AND zone_risk = high THEN suggest: shift to Zone B."

4. Data Availability

Data	Source	Access	Notes
Sentinel-2 imagery	Copernicus Hub	FREE	10m, 5-day revisit
Landsat 8/9	USGS Earth Explorer	FREE	30m resolution
NDVI, soil moisture, temp	Google Earth Engine	FREE	Pre-processed, API
Rainfall data	IMD	FREE	District-level daily
Geological maps	GSI Bhukosh Portal	FREE	Digitized maps
MOIL mine locations	Wikipedia + MOIL site	FREE	All 10 mines documented
MOIL production data	Annual Reports	FREE	Yearly aggregate
Drill-level geological	MOIL internal	N/A	Proprietary
Equipment sensor data	MOIL internal	N/A	Proprietary

Key insight: You have enough public data for satellite + weather + reserve mapping. For production/equipment data, simulate realistic synthetic data based on MOIL's published annual reports.

5. Published Research You Can Build On

There's already a peer-reviewed paper on MOIL's own mine:

"Multispectral Imaging for Characterization of Waste Rock Dumps: A Case Study of Dongri Buzurg Manganese Mine, Maharashtra, India"

- Used Landsat 9 OLI bands for spectral signature analysis
- Identified sensitive bands for manganese-bearing rocks
- Validated with XRF/XRD mineral analysis
- Published in Journal of Geological Society of India

"Earth Observation Approach for Targeting Stratiform Deposit of Manganese in Central India"

- Used ASTER + Sentinel-2B over Balaghat district (MOIL territory)
- Band ratios + PCA for mineralization mapping

You can cite these papers and use their methodology directly. This is your domain knowledge shortcut.

6. What Ministry of Steel Judges Will Look For

They WANT	They DON'T Want
Working dashboard with real satellite imagery of MOIL mine areas	Generic "AI platform" mockup
Actual Sentinel-2/Landsat data loaded for Balaghat/Nagpur	Stock satellite images
Production forecast that responds to weather changes	Static charts
Specific knowledge of MOIL's mines (names, locations, output)	"Some mining company in India"
Corrective actions tied to real mining operations	Vague "AI recommends optimization"
Understanding of monsoon impact on open-cast vs underground mines	Ignoring weather-mining connection

7. Risk Matrix

Risk	Severity	Mitigation
No access to MOIL internal data	HIGH	Use published annual report figures + synthetic data. Be upfront: "With real MOIL data, accuracy improves to X%"
Mineral mapping from satellite too complex	MEDIUM	Focus 70% on production prediction, 30% on reserve mapping. Use the Dongri Buzurg paper's methodology.
Judges are domain experts	HIGH	Learn MOIL mine names, output figures, geological context (Sausar Group, Precambrian stratigraphy).

Risk	Severity	Mitigation
Other teams pick this PS	LOW	Mining + geology is niche. Most teams avoid it.
Satellite data processing takes too long	MEDIUM	Use Google Earth Engine (pre-processed). Don't download raw and process locally.

8. Honest Verdict

Aspect	Rating	Notes
Is the problem real?	YES	MOIL literally wrote this PS. They need this.
Can you win with it?	YES, IF...	...you do the domain prep (mine names, geology basics, published papers)
Too hard to prototype?	NO	Production prediction is standard ML. Satellite side has published methodology.
Low competition?	YES	Mining + geology + space tech = very niche
Saves real money?	YES	MOIL produces 18 lakh tonnes/yr. Even 2% better planning = crores saved.
Biggest trap	—	Spending all 36 hours on satellite mineral detection and ignoring production shortfall prediction (more impactful + buildable)

Final score after deep research: **21/25** — bumped back up from 19 because the Dongri Buzurg published paper and the realization that satellite data is for weather/operational factors (not just mineral detection) make this significantly more feasible than initially thought.

Appendix: MOIL Mine Details

MOIL Limited operates **10 manganese ore mines** — 7 underground and 3 opencast — in Maharashtra (Nagpur/Bhandara districts) and Madhya Pradesh (Balaghat district). The Balaghat mine is the deepest at 660m shaft depth. MOIL produces ~18 lakh tonnes/year and contributes 52% of India's total manganese output.

Mine	State	Type	Notes
Balaghat	Madhya Pradesh	Underground	Largest, deepest (660m)
Ukwa	Madhya Pradesh	Underground	
Tirodi	Madhya Pradesh	Opencast	
Dongri Buzurg	Maharashtra	Opencast	~300K MT/yr. Research paper site.
Kandri	Maharashtra	Underground	
Munsar	Maharashtra	Underground	
Beldongri	Maharashtra	Underground	
Gumgaon	Maharashtra	Underground	

Mine	State	Type	Notes
Chikla	Maharashtra	Underground	
Sitapatore	Maharashtra	Opencast	

Sources

- [MOIL Official Website — moil.nic.in](https://moil.nic.in)
- [Dongri Buzurg Multispectral Study — Springer \(J. Geological Society of India\)](#)
- [Earth Observation for Manganese in Central India — ScienceDirect](#)
- [GSI Geological Data Portal \(Bhukosh\) — ndem.nrsc.gov.in](https://ndem.nrsc.gov.in)
- [MOIL FY25-26 Production Record — The Machine Maker](#)
- [Global Manganese Production Decline 2024 — Mining Technology](#)